

Jiali Shen

University of Helsinki | jiali.shen@Helsinki.fi | +358 505560436 | <https://orcid.org/0000-0001-8701-7929>

<https://researchportal.helsinki.fi/en/persons/jiali-shen> | <https://github.com/Jialishen> | China

RESEARCH PROFILE

A dedicated atmospheric scientist specialising in marine new particle formation and atmospheric chemistry, with a strong record of international collaboration through CLOUD at CERN and multiple field campaigns. My research focuses on the atmospheric oxidation of marine-emitted and biogenic precursors, and their role in particle formation, using a combination of laboratory experiments, kinetic modelling, and field observations.

Education

University of Helsinki, Finland • PhD in Atmospheric Science (Distinction)	2018.05 – 2023.04
Zhejiang University, China • Master in Energy & Environmental Engineering (Exemption graduate)	2015.09 – 2018.03
Zhejiang University, China • Bachelor in Renewable Energy Science & Engineering	2011.09 – 2015.09

Professional Experience

Postdoc researcher , Institute for Atmospheric and Earth system research/Helsinki Institute of Physics, University of Helsinki, Finland	2023.05 – present
Run coordinator , CLOUD project coordinator for marine green runs, University of Helsinki and CERN	2022.04 – 2022.05 2023.10 – 2023.11
Project coordinator , project of Research and Maintenance of Air Quality during the G20 Hangzhou Summit, Zhejiang University	2015.07 – 2016.09

Awards

National Scholarship for Graduate Excellence (1%), China	2018
First Class National Prize of University Contest on Energy Saving and Emission Reduction (1%), China	2014

Fellowship

Marie Skłodowska-Curie Early Stage Researcher Fellow (CLOUD-MOTION ITN), University of Helsinki	2018 - 2024
---	-------------

PhD Training Highlights

- Participated annually in CLOUD experiments at CERN (3 months/year) under the Marie Skłodowska-Curie ITN CLOUD-MOTION.
- Attended over 5 international summer/winter schools and workshops (PSI, Wuppertal, Stockholm, Frankfurt).
- Conducted field observations in Izaña Observatory, Spain, and Qvidja, Finland; completed a secondment at Aerodyne Research Inc.

Key Publications

Summary: Includes 4 first-author papers (including a Nature cover article), 1 co-first-author paper, and 1 corresponding-author paper.

1. J. Shen*, D.M. Russell*, J. DeVivo, . . . , J. Kirkby[#], J. Curtius[#], X.-C. He[#], New particle formation from isoprene under upper tropospheric conditions. **Nature**, (2024).
2. X.-C. He[#], J. Shen[#], S. Iyer, P. Juuti, J. Zhang, M. Koirala, et al., Characterisation of the multi-scheme chemical

ionisation inlet-2 and the detection of gaseous iodine species. **Atmospheric Measurement Techniques**, (2023).

3. W. Scholz*, J. Shen*, D. Aliaga, C. Wu, S. Carbone, I. Moreno et al. Measurement report: Long-range transport and the fate of dimethyl sulfide oxidation products in the free troposphere derived from observations at the high-altitude research station Chacaltaya (5240 m a.s.l.) in the Bolivian Andes. **Atmospheric Chemistry and Physics**, (2023)

4. J. Shen*, W. Scholz, X.-C. He, P. Zhou, G. Marie, M. Wang, et al. High Gas-Phase Methanesulfonic Acid Production in the OH-Initiated Oxidation of Dimethyl Sulfide at Low Temperatures. **Environmental Science & Technology**, (2022).

5. J. Shen*, A. Bigi, A. Marinoni, J. Lampilahti, J. Kontkanen, G. Ciarelli, et al. Emerging Investigator Series: COVID-19 lockdown effects on aerosol particle size distributions in northern Italy. **Environmental Science: Atmospheres**, (2021)

6. J. Shen*, C. Zheng, L. Xu, Y Zhang, Y. Zhang, S. Liu, X. Gao. Atmospheric emission inventory of SO₃ from coal-fired power plants in China in the period 2009–2014. **Atmospheric Environment**, (2019)

*Co-first; #Corresponding authors

Presentations

European Aerosol Conference (2021 Talk, 2024 session co-chair & Talk)

International Conference on Nucleation and Atmospheric Aerosols (2023 Talk)

Hvittorp workshop of VILMA (2022 Talk)

CLOUD workshops (2018-2024 Talk)

Invited Seminar: Zhejiang University (2023.08)

Teaching Experience

University of Helsinki

Polar Oceans and Ice (2025), 5 ECTS

Guest Lecture & TA

Laboratory Aerosol Instrument Measurement (2024),5 ECTS

Co-instructor (1 of 3)

Formation and Growth of Atmospheric Aerosols (2024), 5 ECTS

Lead TA & Organizer

Advanced Analysis of Atmosphere–Surface Interactions and Feedbacks (2020)

Teaching assistant

Pedagogical Training

- **Learning in higher education**, University of Helsinki, (5 ECTS, 2024)
- **Constructive Alignment in Course Design**, University of Helsinki, (5 ECTS, 2024)
- **Assessment of learning and giving feedback**, University of Helsinki, (5 ECTS, 2024)
- **Academic Supervision and Supervisor Training**, University of Helsinki, (5 ECTS, 2025)

Outreach & Professional Development

Peer Reviewer: Atmospheric Chemistry and Physics, Environmental Science & Technology

Media Coverage: Advanced Science News, 2024, Scientists identify the driver behind a mysterious particle cloud over the Amazon

Supervision & Mentoring

Wenjuan Yu, Mentor for her doctoral study

2023.02 - present

Full publications

1. Caudillo, L., Rörup, B., Heinritzi, M., Marie, G., Simon, M., Wagner, A. C., Müller, T., Granzin, M., Amorim, A., Ataei, F., Baalbaki, R., Bertozzi, B., Brasseur, Z., Chiu, R., Chu, B., Dada, L., Duplissy, J., Finkenzeller, H., Gonzalez Carracedo, L., . . . Curtius, J. (2021). Chemical composition of nanoparticles from α -pinene nucleation and the influence of isoprene and relative humidity at low temperature. *Atmospheric Chemistry and Physics*, 21(22), 17099–17114. <https://doi.org/10.5194/acp-21-17099-2021>
2. Caudillo, L., Surdu, M., Lopez, B., Wang, M., Thoma, M., Bräkling, S., Buchholz, A., Simon, M., Wagner, A. C., Müller, T., Granzin, M., Heinritzi, M., Amorim, A., Bell, D. M., Brasseur, Z., Dada, L., Duplissy, J., Finkenzeller, H., He, X.-C., . . . Curtius, J. (2022). An intercomparison study of four different techniques for measuring the chemical composition of nanoparticles. *Atmospheric Chemistry and Physics*, 1–24. <https://doi.org/10.5194/acp-2022-498>
3. Ciarelli, G., Jiang, J., Haddad, I. E., Bigi, A., Aksoyoglu, S., Prévôt, A. S. H., Marinoni, A., Shen, J., Yan, C., & Bianchi, F. (2021). Modeling the effect of reduced traffic due to COVID-19 measures on air quality using a chemical transport model: Impacts on the Po Valley and the Swiss Plateau regions. *Environ. Sci.: Atmos.*, 1(5), 228–240. <https://doi.org/10.1039/D1EA00036E>
4. Dada, L., Okuljar, M., Shen, J., Olin, M., Wu, Y., Heimsch, L., Herlin, I., Kankaanrinta, S., Lampimäki, M., Kalliokoski, J., Baalbaki, R., Lohila, A., Petäjä, T., Maso, M. D., Duplissy, J., Kerminen, V.-M., & Kulmala, M. (2023). The synergistic role of sulfuric acid, ammonia and organics in particle formation over an agricultural land. *Environ. Sci.: Atmos.*, 3(8), 1195–1211. <https://doi.org/10.1039/D3EA00065F>
5. Finkenzeller, H., Iyer, S., He, X.-C., Simon, M., Koenig, T. K., Lee, C. F., Valiev, R., Hofbauer, V., Amorim, A., Baalbaki, R., Baccarini, A., Beck, L., Bell, D. M., Caudillo, L., Chen, D., Chiu, R., Chu, B., Dada, L., Duplissy, J., . . . Volkamer, R. (2023). The gas-phase formation mechanism of iodic acid as an atmospheric aerosol source. *Nat. Chem.*, 15(1), 129–135. <https://doi.org/10.1038/s41557-022-01067-z>
6. He, X.-C., Iyer, S., Sipilä, M., Ylisirniö, A., Peltola, M., Kontkanen, J., Baalbaki, R., Simon, M., Kürten, A., Tham, Y. J., Pesonen, J., Ahonen, L. R., Amanatidis, S., Amorim, A., Baccarini, A., Beck, L., Bianchi, F., Brilke, S., Chen, D., . . . Kulmala, M. (2021). Determination of the collision rate coefficient between charged iodic acid clusters and iodic acid using the appearance time method. *Aerosol Science and Technology*, 55(2), 231–242. <https://doi.org/10.1080/02786826.2020.1839013>
7. He, X.-C., Shen, J., Iyer, S., Juuti, P., Zhang, J., Koirala, M., Kytökari, M. M., Worsnop, D. R., Rissanen, M., Kulmala, M., Maier, N. M., Mikkilä, J., Sipilä, M., & Kangasluoma, J. (2023). Characterisation of gaseous iodine species detection using the multi-scheme chemical ionisation inlet 2 with bromide and nitrate chemical ionisation methods. *Atmospheric Measurement Techniques*, 16(19), 4461–4487. <https://doi.org/10.5194/amt-16-4461-2023>
8. He, X.-C., Simon, M., Iyer, S., Xie, H.-B., Rörup, B., Shen, J., Finkenzeller, H., Stolzenburg, D., Zhang, R., Baccarini, A., Tham, Y. J., Wang, M., Amanatidis, S., Piedehierro, A. A., Amorim, A., Baalbaki, R., Brasseur, Z., Caudillo, L., Chu, B., . . . Kulmala, M. (2023). Iodine oxoacids enhance nucleation of sulfuric acid particles in the atmosphere. *Science*, 382(6676), 1308–1314. <https://doi.org/10.1126/science.adh2526>
9. He, X.-C., Tham, Y. J., Dada, L., Wang, M., Finkenzeller, H., Stolzenburg, D., Iyer, S., Simon, M., Kürten, A., Shen, J., Rörup, B., Rissanen, M., Schobesberger, S., Baalbaki, R., Wang, D. S., Koenig, T. K., Jokinen, T., Sarnela, N., Beck, L. J., . . . Sipilä, M. (2021). Role of iodine oxoacids in atmospheric aerosol nucleation. *Science*, 371(6529), 589–595. <https://doi.org/10.1126/science.abe0298>
10. Li, D., Huang, W., Wang, D., Wang, M., Thornton, J. A., Caudillo, L., Rörup, B., Marten, R., Scholz, W., Finkenzeller, H., Marie, G., Baltensperger, U., Bell, D. M., Brasseur, Z., Curtius, J., Dada, L., Duplissy, J., Gong, X., Hansel, A., . . . Riva, M. (2024). Nitrate Radicals Suppress Biogenic New Particle Formation from Monoterpene Oxidation. *Environ. Sci. Technol.*, 58(3), 1601–1614. <https://doi.org/10.1021/acs.est.3c07958>
11. Marten, R., Xiao, M., Rörup, B., Wang, M., Kong, W., He, X.-C., Stolzenburg, D., Pfeifer, J., Marie, G., Wang, D. S., Scholz, W., Baccarini, A., Lee, C. P., Amorim, A., Baalbaki, R., Bell, D. M., Bertozzi, B., Caudillo, L., Chu, B., . . . Haddad, I. E. (2022). Survival of newly formed particles in haze conditions. *Environ. Sci.: Atmos.*, 2(3), 491–499. <https://doi.org/10.1039/D2EA00007E>

12. Marten, R., Xiao, M., Wang, M., Kong, W., He, X.-C., Stolzenburg, D., Pfeifer, J., Marie, G., Wang, D. S., Elser, M., Baccarini, A., Lee, C. P., Amorim, A., Baalbaki, R., Bell, D. M., Bertozzi, B., Caudillo, L., Dada, L., Duplissy, J., ... El Haddad, I. (2024). Assessing the importance of nitric acid and ammonia for particle growth in the polluted boundary layer. *Environ. Sci.: Atmos.*, 4(2), 265–274. <https://doi.org/10.1039/D3EA00001J>
13. Olin, M., Okuljar, M., Rissanen, M. P., Kalliokoski, J., Shen, J., Dada, L., Lampimäki, M., Wu, Y., Lohila, A., Duplissy, J., Sipilä, M., Petäjä, T., Kulmala, M., & Dal Maso, M. (2022). Measurement report: Atmospheric new particle formation in a coastal agricultural site explained with binPMF analysis of nitrate CI-API-TOF spectra. *Atmospheric Chemistry and Physics*, 22(12), 8097–8115. <https://doi.org/10.5194/acp-22-8097-2022>
14. Rörup, B., He, X.-C., Shen, J., Baalbaki, R., Dada, L., Sipilä, M., Kirkby, J., Kulmala, M., Amorim, A., Baccarini, A., Bell, D. M., Caudillo-Plath, L., Duplissy, J., Finkenzeller, H., Kürten, A., Lamkaddam, H., Lee, C. P., Makhmutov, V., Manninen, H. E., ... Lehtipalo, K. (2024). Temperature, humidity, and ionisation effect of iodine oxoacid nucleation. *Environ. Sci.: Atmos.*, 4(5), 531–546. <https://doi.org/10.1039/D4EA00013G>
15. Scholz, W., Shen, J., Aliaga, D., Wu, C., Carbone, S., Moreno, I., Zha, Q., Huang, W., Heikkinen, L., Jaffrezo, J. L., Uzu, G., Partoll, E., Leiminger, M., Velarde, F., Laj, P., Ginot, P., Artaxo, P., Wiedensohler, A., Kulmala, M., ... Hansel, A. (2023). Measurement report: Long-range transport and the fate of dimethyl sulfide oxidation products in the free troposphere derived from observations at the high-altitude research station Chacaltaya (5240 a.s.l.) in the Bolivian Andes. *Atmospheric Chemistry and Physics*, 23(2), 895–920. <https://doi.org/10.5194/acp-23-895-2023>
16. Shen, J., Bigi, A., Marinoni, A., Lampilahti, J., Kontkanen, J., Ciarelli, G., Putaud, J. P., Nieminen, T., Kulmala, M., Lehtipalo, K., & Bianchi, F. (2021). Emerging Investigator Series: COVID-19 lockdown effects on aerosol particle size distributions in northern Italy. *Environ. Sci.: Atmos.*, 1(5), 214–227. <https://doi.org/10.1039/D1EA00016K>
17. Shen, J., Russell, D. M., DeVivo, J., Kunkler, F., Baalbaki, R., Mentler, B., Scholz, W., Yu, W., Caudillo-Plath, L., Sommer, E., Ahongshangbam, E., Alfaouri, D., Almeida, J., Amorim, A., Beck, L. J., Beckmann, H., Berntheusel, M., Bhattacharyya, N., Canagaratna, M. R., ... He, X.-C. (2024). New particle formation from isoprene under upper-tropospheric conditions. *Nature*, 636(8041), 115–123. <https://doi.org/10.1038/s41586-024-08196-0>
18. Shen, J., Scholz, W., He, X.-C., Zhou, P., Marie, G., Wang, M., Marten, R., Surdu, M., Rörup, B., Baalbaki, R., Amorim, A., Ataei, F., Bell, D. M., Bertozzi, B., Brasseur, Z., Caudillo, L., Chen, D., Chu, B., Dada, L., ... Worsnop, D. R. (2022). High Gas-Phase Methanesulfonic Acid Production in the OH-Initiated Oxidation of Dimethyl Sulfide at Low Temperatures. *Environ. Sci. Technol.*, 56(19), 13931–13944. <https://doi.org/10.1021/acs.est.2c05154>
19. Shen, J., Zheng, C., Xu, L., Zhang, Y., Zhang, Y., Liu, S., & Gao, X. (2019). Atmospheric emission inventory of SO₃ from coal-fired power plants in China in the period 2009–2014. *Atmospheric Environment*, 197, 14–21. <https://doi.org/10.1016/j.atmosenv.2018.10.008>
20. Simon, M., Dada, L., Heinritzi, M., Scholz, W., Stolzenburg, D., Fischer, L., Wagner, A. C., Kürten, A., Rörup, B., He, X.-C., Almeida, J., Baalbaki, R., Baccarini, A., Bauer, P. S., Beck, L., Bergen, A., Bianchi, F., Bräkling, S., Brilke, S., ... Curtius, J. (2020). Molecular understanding of new-particle formation from α -pinene between 50 and +25 °C. *Atmospheric Chemistry and Physics*, 20(15), 9183–9207. <https://doi.org/10.5194/acp-20-9183-2020>
21. Stolzenburg, D., Simon, M., Ranjithkumar, A., Kürten, A., Lehtipalo, K., Gordon, H., Ehrhart, S., Finkenzeller, H., Pichelstorfer, L., Nieminen, T., He, X.-C., Brilke, S., Xiao, M., Amorim, A., Baalbaki, R., Baccarini, A., Beck, L., Bräkling, S., Caudillo Murillo, L., ... Winkler, P. M. (2020). Enhanced growth rate of atmospheric particles from sulfuric acid. *Atmospheric Chemistry and Physics*, 20(12), 7359–7372. <https://doi.org/10.5194/acp-20-7359-2020>
22. Surdu, M., Lamkaddam, H., Wang, D. S., Bell, D. M., Xiao, M., Lee, C. P., Li, D., Caudillo, L., Marie, G., Scholz, W., Wang, M., Lopez, B., Piedehierro, A. A., Ataei, F., Baalbaki, R., Bertozzi, B., Bogert, P., Brasseur, Z., Dada, L., ... El Haddad, I. (2023). Molecular Understanding of the Enhancement in Organic Aerosol Mass at High Relative Humidity. *Environ. Sci. Technol.*, 57(6), 2297–2309. <https://doi.org/10.1021/acs.est.2c04587>

23. Tham, Y. J., He, X.-C., Li, Q., Cuevas, C. A., Shen, J., Kalliokoski, J., Yan, C., Iyer, S., Lehmusjärvi, T., Jang, S., Thakur, R. C., Beck, L., Kemppainen, D., Olin, M., Sarnela, N., Mikkilä, J., Hakala, J., Marbouti, M., Yao, L., . . . Sipilä, M. (2021). Direct field evidence of autocatalytic iodine release from atmospheric aerosol. *Proceedings of the National Academy of Sciences*, 118(4), e2009951118. <https://doi.org/10.1073/pnas.2009951118>
24. Wang, M., He, X.-C., Finkenzeller, H., Iyer, S., Chen, D., Shen, J., Simon, M., Hofbauer, V., Kirkby, J., Curtius, J., Maier, N., Kurtén, T., Worsnop, D. R., Kulmala, M., Rissanen, M., Volkamer, R., Tham, Y. J., Donahue, N. M., & Sipilä, M. (2021). Measurement of iodine species and sulfuric acid using bromide chemical ionization mass spectrometers. *Atmospheric Measurement Techniques*, 14(6), 4187–4202. <https://doi.org/10.5194/amt-14-4187-2021>
25. Wang, M., Kong, W., Marten, R., He, X.-C., Chen, D., Pfeifer, J., Heitto, A., Kontkanen, J., Dada, L., Kürten, A., Yli-Juuti, T., Manninen, H. E., Amanatidis, S., Amorim, A., Baalbaki, R., Baccarini, A., Bell, D. M., Bertozzi, B., Bräkling, S., . . . Donahue, N. M. (2020). Rapid growth of new atmospheric particles by nitric acid and ammonia condensation. *Nature*, 581(7807), 184–189. <https://doi.org/10.1038/s41586-020-2270-4>
26. Wang, M., Xiao, M., Bertozzi, B., Marie, G., Rörup, B., Schulze, B., Bardakov, R., He, X.-C., Shen, J., Scholz, W., Marten, R., Dada, L., Baalbaki, R., Lopez, B., Lamkaddam, H., Manninen, H. E., Amorim, A., Ataei, F., Bogert, P., . . . Donahue, N. M. (2022). Synergistic HNO_3 – H_2SO_4 – NH_3 upper tropospheric particle formation. *Nature*, 605(7910), 483–489. <https://doi.org/10.1038/s41586-022-04605-4>
27. Yao, L., Fan, X., Yan, C., Kurtén, T., Daellenbach, K. R., Li, C., Wang, Y., Guo, Y., Dada, L., Rissanen, M. P., Cai, J., Tham, Y. J., Zha, Q., Zhang, S., Du, W., Yu, M., Zheng, F., Zhou, Y., Kontkanen, J., . . . Bianchi, F. (2020). Unprecedented Ambient Sulfur Trioxide (SO_3) Detection: Possible Formation Mechanism and Atmospheric Implications. *Environ. Sci. Technol. Lett.*, 7(11), 809–818. <https://doi.org/10.1021/acs.estlett.0c00615>
28. Zhang, R., Xie, H.-B., Ma, F., Chen, J., Iyer, S., Simon, M., Heinritzi, M., Shen, J., Tham, Y. J., Kurtén, T., Worsnop, D. R., Kirkby, J., Curtius, J., Sipilä, M., Kulmala, M., & He, X.-C. (2022). Critical Role of Iodous Acid in Neutral Iodine Oxoacid Nucleation. *Environ. Sci. Technol.*, 56(19), 14166–14177. <https://doi.org/10.1021/acs.est.2c04328>
29. Zheng, C., Shao, L., Wang, Y., Zheng, H., Gao, W., Zhang, H., Wu, Z., Shen, J., & Gao, X. (2021). Investigation of the growth and removal of particles in coal-fired flue gas by temperature management. *Fuel*, 302, 121220. <https://doi.org/10.1016/j.fuel.2021.121220>
30. Zheng, C., Shen, J., Zhang, Y., Huang, W., Zhu, X., Wu, X., Chen, L., Gao, X., & Cen, K. (2017). Quantitative assessment of industrial VOC emissions in China: Historical trend, spatial distribution, uncertainties, and projection. *Atmospheric Environment*, 150, 116–125. <https://doi.org/10.1016/j.atmosenv.2016.11.023>
31. Zheng, C., Shen, J., Zhang, Y., Zhu, X., Wu, X., Chen, L., & Gao, X. (2017). Atmospheric Emission Characteristics and Control Policies of Anthropogenic VOCs from Industrial Sources in Yangtze River Delta Region, China. *Aerosol Air Qual. Res.*, 17(9), 2263–2275. <https://doi.org/10.4209/aaqr.2016.06.0234>
32. Zheng, C., Zheng, H., Shen, J., Gao, W., Yang, Z., Zhao, Z., Wang, Y., Zhang, H., & Gao, X. (2020). Evolution of Condensable Fine Particle Size Distribution in Simulated Flue Gas by External Regulation for Growth Enhancement. *Environ. Sci. Technol.*, 54(7), 3840–3848. <https://doi.org/10.1021/acs.est.9b06569>